AI Agents for Personalized Health Coaching

A Multi-Agent Generative AI Architecture with Deterministic Health Calculation Engines

Course: Generative AI (GenAI) — TAE 1	**Framework: Python, Streamlit, LLM APIs**	**Architecture: Coordinator-Subagent Multi-Agent**

1. Introduction

1.1 Background & Motivation

Modern health and wellness management is inherently multi-faceted, requiring coordinated insights across clinical nutrition, physiological exercise science, and behavioral lifestyle recovery. Generic conversational Large Language Models (LLMs) often struggle in personalized wellness guidance due to a lack of domain compartmentalization, arithmetic inaccuracy in calculating metabolic requirements (such as BMR, TDEE, and macronutrient targets), and the generation of conflicting or generic advice without strict constraint enforcement.

To overcome these limitations, this project introduces a **Hybrid Multi-Agent Generative AI Framework** specifically engineered for personalized health coaching. By pairing a deterministic mathematical engine with autonomous domain-specialized agents (Nutrition, Fitness, and Lifestyle) under a Master Coordinator Agent, the system provides mathematically grounded, hyper-personalized, and safe daily health recommendations.

1.2 Problem Statement

Monolithic single-agent chatbots present severe vulnerabilities when applied to wellness coaching:

- **Domain Hallucination:** Unspecialized LLMs intermingle calorie calculation with exercise splits, leading to mathematically invalid diet regimens and dangerous macro ratios.
- **Lack of Contextual Grounding:** Off-the-shelf chatbots fail to persistently respect user constraints (such as vegetarian/vegan dietary habits, metabolic baselines, and age limitations).
- **Single Point of Failure:** Unstructured prompting lacks modular fallback mechanisms when external LLM API rate limits or network latencies occur.

1.3 Key Objectives & Scope

The primary objectives of this GenAI TAE 1 project include:

1. **Multi-Agent Orchestration:** Formulate an asynchronous Coordinator-Subagent architecture that intelligently parses user queries and dispatches sub-tasks to dedicated specialist agents.
2. **Deterministic + Stochastic Hybridization:** Integrate scientific physiological formulas (Mifflin-St Jeor equation, TDEE activity multipliers, BMI classification) to ground LLM reasoning in verified clinical mathematics.
3. **Dual-Engine LLM Fallback:** Enable seamless execution with high-performance LLMs (Google Gemini 2.5/1.5 Flash, Groq LLaMA-3) along with an intelligent offline rule-based fallback engine.
4. **Interactive Clinical Dashboard:** Implement an intuitive, reactive Streamlit web interface with real-time biometric metrics, dynamic agent badges, and persistent state management.

2. Methodology

2.1 System Architecture & Data Pipeline

The system employs a hierarchical **Coordinator-Subagent Pattern**. When a user submits an inquiry alongside their profile metadata (Age, Gender, Height, Weight, Goal, Diet Preference, Activity Level), the input is processed through a sequential two-stage pipeline:

Component Layer	Module / File	Functional Responsibility
1. Deterministic Engine	`utils/health_calc.py`	Calculates BMI, BMR (Mifflin-St Jeor), TDEE, caloric shifts (-400/+300 kcal), and water intake.
2. Master Coordinator	`agents/coordinator_agent.py`	Semantic intent routing, agent activation filtering, and multi-agent response synthesis.
3. Nutrition Specialist	`agents/nutrition_agent.py`	Macro distributions, meal plans (Veg/Vegan/Non-Veg), and micronutrient timing.
4. Fitness Specialist	`agents/fitness_agent.py`	Resistance/Cardio training splits, sets/reps, warm-up, and recovery protocols.
5. Lifestyle Specialist	`agents/lifestyle_agent.py`	Circadian alignment, sleep hygiene protocols, hydration schedules, and box-breathing.
6. Storage & State	`utils/storage.py`	Persistent JSON serialization of user profile biometric states and conversational history.

2.2 Deterministic Health Calculation Formulas

To prevent arithmetic inaccuracies common in standard LLMs, biometric calculations are computed using exact clinical formulas before being injected into prompt contexts:

- **Basal Metabolic Rate (BMR) [Mifflin-St Jeor Equation]:**

  Men: $\mathrm{BMR} = (10 \times \mathrm{weight}_{kg}) + (6.25 \times \mathrm{height}_{cm}) - (5 \times \mathrm{age}) + 5$
  Women: $\mathrm{BMR} = (10 \times \mathrm{weight}_{kg}) + (6.25 \times \mathrm{height}_{cm}) - (5 \times \mathrm{age}) - 161$
- **Total Daily Energy Expenditure (TDEE):** TDEE = BMR × Activity Multiplier [1.2 (Sedentary) to 1.9 (Extra Active)].
- **Caloric & Macro Targets:** Weight Loss applies a 400 kcal deficit (35% Protein, 40% Carb, 25% Fat); Muscle Gain applies a +300 kcal surplus (30% Protein, 45% Carb, 25% Fat).

2.3 Coordinator Dispatch Logic & Prompt Engineering

The Coordinator Agent evaluates incoming user prompts against semantic keyword patterns. Queries containing diet terms trigger the Nutrition Agent; exercise queries activate the Fitness Agent; sleep/hydration queries trigger the Lifestyle Agent. General queries simultaneously invoke all three sub-agents for comprehensive holistic coaching. Each agent is governed by strict system prompts enforcing safety constraints, dietary compliance, and explicit non-medical disclaimers.

3. Result (Work Done)

3.1 Implemented Artifacts & User Interface

The full multi-agent application was implemented, tested, and validated. The frontend features a sleek dark-themed **Glassmorphism UI** built with Streamlit and modern CSS typography (`Outfit`), offering high interactivity and visual clarity:

- **Sidebar Biometric Onboarding:** Users input personal metrics (Age, Gender, Height, Weight, Activity, Diet, Goal, API Key) with instant real-time metric updates.
- **Live Biometric Metric Cards:** Displays BMI badge with health category, BMR, TDEE, Calorie Goal, and Daily Hydration target in Liters.
- **Interactive Multi-Agent Chat Console:** Displays transparent agent badges (■■ Nutrition Agent, ■■ Fitness Agent, ■ Lifestyle Agent) showing which agents contributed to each response.

3.2 Sample Execution & Multi-Agent Collaboration Trace

Input User Query: *"I am a 22-year-old vegetarian wanting to build lean muscle. Give me a full daily plan."*
Orchestration Result: Coordinator identified cross-domain requirements and activated all 3 Specialist Agents.
• **Nutrition Agent:** Calculated 2,450 kcal (+300 surplus), 183g Protein (Paneer, Tofu, Greek Yogurt, Dal, Whey), 275g Carbs, 68g Healthy Fats with timed meal schedules.
• **Fitness Agent:** 4-day Hypertrophy split (Compound Push/Pull/Legs), 4 sets x 8-12 reps with 90s rest, warm-up and cool-down routines.
• **Lifestyle Agent:** 3.4L hydration schedule, 8-hour sleep hygiene with 45-min blue light cutoff, and 4-4-4-4 box breathing.

3.3 Evaluation & Key Performance Highlights

Evaluation Metric	Standard Generic Chatbot	Proposed Multi-Agent System
Macro Accuracy	Approximated / Inconsistent arithmetic	100% Grounded in Mifflin-St Jeor & exact grams
Dietary Adherence	Frequently hallucinates non-veg items	Strict filtering strictly adhering to user diet
Offline Resilience	Fails completely without active API	Smart deterministic rule fallback engine
Safety Protocols	Vague or missing warnings	Enforced non-medical clinical disclaimer

3.4 Conclusion & Future Work

The project successfully demonstrates the effectiveness of multi-agent collaborative architectures for specialized Generative AI applications. By decoupling clinical nutrition, strength coaching, and behavioral lifestyle recovery into dedicated sub-agents backed by deterministic mathematical verification, the system achieves superior accuracy, personalization, and safety compared to conventional single-agent LLMs. Future extensions include wearable IoT integration (smartwatch heart-rate sync), continuous vision-based food meal calorie scanning, and long-term reinforcement learning from user feedback.